

Dynamic Pricing under Capacity Constraints: An Agent-Based Approach to Competition in the Car Rental Industry

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Outline

- ① Motivation and objectives
- ② Theoretical framework and agent-based model
- ③ Experimental design
- ④ Results
- ⑤ Conclusions and limitations

Motivation and Context

- The car rental market combines three key characteristics:
 - **fixed capacity** in the short run,
 - **perishable** inventory,
 - competitor prices **observable in real time**.
- Pricing decisions often incorporate heuristic rules alongside formal optimization models.
- During an internship in a *Pricing & Revenue* department, a key question emerged:

Motivating Question

Is tracking one's relative position against close competitors more effective than relying on aggregate market indicators?

Objective and Research Questions

Objective

Evaluate, via an Agent-Based Model, whether a pricing rule based on relative competitive positioning improves performance compared to rules based on market averages or own occupancy.

Three main questions:

- ❶ **RQ1:** Does the Positional strategy yield higher RevPAR than the Average strategy?
- ❷ **RQ2:** Does the Positional strategy reduce occupancy instability compared to the Occupancy strategy?
- ❸ **RQ3:** Does combining competitive information and occupancy enhance performance through a Hybrid strategy?

Additionally, the effect of price sensitivity is explored via a parameter sweep over β .

Theoretical Framework

Competition

Bertrand–Edgeworth

Capacity constraints prevent standard Bertrand competitive outcomes from applying directly.

Key idea: Capacity bounds the volume each firm can serve; hence, price and availability are interdependent.

Revenue Management

Perishable Inventory

Available capacity is fixed in the short run, and an unsold unit represents forfeited revenue.

Key idea: Pricing must adapt dynamically to the tension among demand, price, and remaining capacity.

Agent-Based Models

Agents + Heuristics

Enables the study of repeated interactions among agents with bounded rationality and decision rules.

Key idea: Facilitates comparing alternative pricing rules under identical market conditions.

Methodological Choice

The ABM bridges the three pillars of the problem: **competition**, **capacity constraints**, and **heuristic decision-making**, allowing for controlled counterfactual experiments.

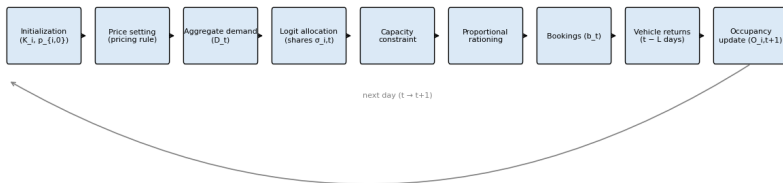
Agent-Based Model Structure

Simulated Market

- $N = 10$ firms.
- Fleet sizes: $K_i \in [80, 150]$ vehicles.
- Time horizon: $T = 180$ days.
- 20 seeds per scenario.
- All firms follow the same rule within each simulated market.

Firm State

- $p_{i,t}$: price of firm i on day t .
- $O_{i,t}$: occupied vehicles.
- $o_{i,t} = \frac{O_{i,t}}{K_i}$: occupancy rate.



Demand and Allocation Mechanism

Aggregate Demand

$$D_t = D_0 s_t (1 + \varepsilon_t)$$

with:

$$s_{\text{low}} = 1.0, \quad s_{\text{high}} = 1.6, \quad \sigma_D = 0.05.$$

Capacity Constraint

$$b_{i,t} = \min(d_{i,t}, K_i - O_{i,t})$$

Unserved demand is redistributed proportionally among firms with remaining capacity.

Allocation Across Firms

Logit model:

$$\sigma_{i,t} = \frac{\exp(-\beta p_{i,t})}{\sum_{j=1}^N \exp(-\beta p_{j,t})}$$

$$d_{i,t} = D_t \sigma_{i,t}$$

Inventory Dynamics

- Rental duration: $L = 4$ days.
- Occupancy updates as rentals commence and return.
- Available capacity bounds demand that translates into realized bookings.

The Four Pricing Strategies (I)

All strategies share the same adjustment rule:

$$p_{i,t+1} = \text{clip} [p_{i,t}(1 + \lambda\Delta_{i,t}), p_{\min}, p_{\max}]$$

with $\lambda = 0.03$.

Average

Compares own price against the **average price of the other 9 competitors**.

$$\Delta^M = \frac{\bar{p}_{-i} - p_i}{\bar{p}_{-i}}$$

Occupancy

Adjusts price based on **own capacity utilization**.

$$\Delta^O = \frac{o_i - o^*}{o^*}$$

with $o^* = 0.85$.

The Four Pricing Strategies (II)

The remaining two strategies follow the same adjustment mechanism introduced on the previous slide.

Positional

Compares against the **3 closest competitors in price**.

$$\Delta^P = \frac{\bar{p}_i^{Top3} - p_i}{\bar{p}_i^{Top3}}$$

Hybrid

Blends occupancy and positional signals:

$$\Delta^H = 0.5\Delta^O + 0.5\Delta^P$$

Two experimental dimensions are combined

1. Capacity Tightness

- **Low season** ($s_t = 1.0$): relatively loose capacity.
- **High season** ($s_t = 1.6$): higher capacity pressure.

2. Fleet Size Dispersion

- **Homogeneous**: similarly sized fleets.
- **Dispersed**: coexistence of large and small operators.

	Homogeneous	Dispersed
Low season	✓	✓
High season	✓	✓

4 **strategies** × 4 **scenarios** × 20 **seeds**

Performance Evaluation

Primary Metric: RevPAR

$$\text{RevPAR} = \frac{\sum_t \sum_i R_{i,t}}{\sum_i K_i T_{\text{effective}}}$$

Measures revenue per available vehicle per day.

The initial 20-day window is excluded as *burn-in*.

Occupancy Stability

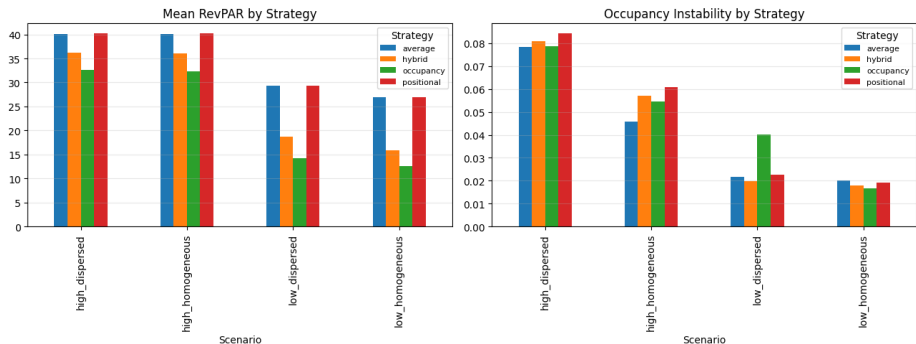
Temporal variability of occupancy is assessed.

Allows examining whether a strategy generates a more stable inventory flow in addition to maximizing revenue.

Statistical Comparison

Paired cross-seed tests with Holm–Bonferroni corrections are applied where appropriate.

Results: Effect of Seasonality and Capacity



Main finding: RevPAR roughly doubles when moving from low to high season, regardless of the pricing strategy employed.

Capacity tightness exerts a substantially larger effect on revenue levels than fleet dispersion does.

RQ1 – Positional vs. Average

Question

Does targeting the 3 closest competitors in price outperform tracking the market average?

Result

NO statistically significant difference is found.

Across all four scenarios, confidence intervals include zero and p -values exceed 0.05.

Scenario	Diff. Average - Positional	p
Low, homogeneous	+0.118	0.070
Low, dispersed	+0.093	0.165
High, homogeneous	−0.110	0.215
High, dispersed	−0.119	0.211

Interpretation

Localized competitive intelligence does not confer a statistically demonstrable advantage over aggregate market signals across the tested scenarios.

RQ1 – Price Sensitivity

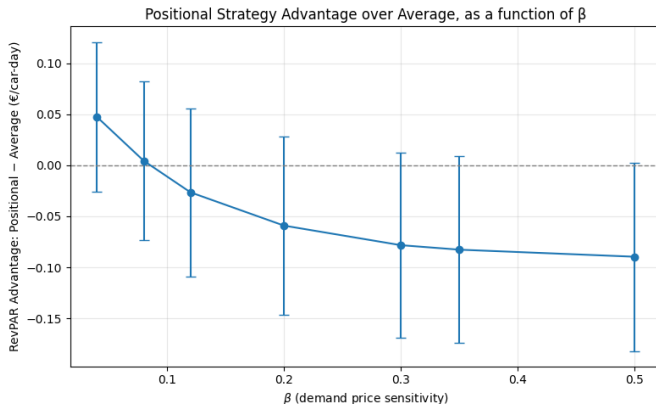
A parameter sweep over price sensitivity was also conducted:

$$\beta \in \{0.04, 0.08, 0.12, 0.20, 0.30, 0.35, 0.50\}$$

Result

No statistically significant difference emerges between Positional and Average across any of the tested values:

$$p \geq 0.061$$



RQ2 – Occupancy Stability

Question

Does the Positional strategy reduce occupancy instability?

Main finding:

- No systematic advantage is observed for the Positional strategy.
- Following Holm–Bonferroni correction, most initially significant differences vanish.
- The only robust patterns involve the Occupancy rule and depend on fleet size dispersion.

Key Takeaway

The preliminary result suggesting higher stability under the Positional rule disappears once the initial 20-day *burn-in* is properly excluded.

RQ3 – Does the Hybrid Strategy Work?

Question

Does blending occupancy and competitive positioning improve outcomes?

Positional $\approx$ Average $>$

$>$ Hybrid $>$ Occupancy

Results:

- Hybrid outperforms Occupancy:

$$p < 0.0001$$

across all four scenarios.

- However, Hybrid performs significantly worse than Positional/Average:

$$p < 0.0001$$

across all relevant pairwise comparisons.

- Under $\gamma = 0.5$, incorporating the occupancy signal dilutes the performance of the positional heuristic.

Conclusions and Implications

1. Positional does not dominate Average

- No statistical evidence supports a systematic edge.

2. Seasonality matters more than the rule

- Higher capacity pressure substantially boosts RevPAR across all strategies.

3. Occupancy stability does not cleanly separate strategies

- Observed effects are contingent on fleet heterogeneity.

4. Hybrid performance is the most robust finding

- A 50/50 mix fails to improve upon pure competitive rules.

Revenue Management Implication

Tracking a small cluster of close rivals represents an operationally transparent heuristic with no noticeable RevPAR penalty relative to monitoring the broad market, yet the model reveals no clear economic superiority for positional targeting.

Limitations and Future Research

Limitations

- Synthetic demand parameters.
- Zero marginal operating costs.
- Deterministic rental duration.
- Proportional demand rationing.
- No firm entry or exit.
- Absence of collusive dynamics.
- Non-optimized $\gamma = 0.5$ weighting.

Extensions

- Incorporate costs and analyze net margins.
- Systematically explore alternative γ values.
- Treat price elasticity as a factorial dimension.
- Vary the number of competing firms.
- Allow endogenous market entry and exit.
- Implement adaptive strategy switching.
- Calibrate with empirical industry data.

Interpretation

Results should be interpreted as **comparative and qualitative evidence**, rather than an exact quantitative forecast of an individual firm's RevPAR.

Thank you

Questions?

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